Huygens' Workshop

toy piano and electronics

Neil Thornock

Program Note

Christiaan Huygens (1629-1695) was an important Dutch mathematician, astronomer, and physicist. Among his many accomplishments, he discovered Saturn's moon Titan and gave the first accurate description of the rings of Saturn. To improve astronomical observations, he also developed the first accurate pendulum clock (achieving accuracy to one minute per day) and later constructed the first accurate spring-regulated clock. He is also credited with one of the earliest theses discussing extraterrestrial life and with a treatise describing 31-tone equal temperament.

I tried to touch on each of these aspects of his career in the two movements of this work. The title *Saturn's Child* refers both to its moon and to the fictive possibility of alien life on Saturn, while the microtonal gestures employ 31-tone equal temperament in a rather cursory nod to his treatise. The music depicts a space explorer orbiting around Saturn, passing through one of its rings (the "stardust" material), coming near collision with Titan, and finally engaging in an austere, transcendent meeting with life from space.

The second movement depicts a clock shop – perhaps an old Renaissance cottage shop in which all the clocks are initially set for the same time, gradually going out of sync, creating pandemonium, and leading to the explosion and demolition of the old cottage – the sole survivor being a trusty grandfather clock, faithfully beating away its inaccurate version of time. The toy piano seemed a perfect instrument for this role: the tinkling sound is reminiscent of dusty, cheap clock bars, and the inaccurate playing mechanism parallels the (by today's standards) inaccurate timekeeping of days gone by.

Performance Notes

Performance requires an amplified toy piano of 1-1/2 octave range (C to F) and Max/MSP. Instructions pertaining to the electronics are included in the main Max patch window.

An audio CD intended for practice purposes will be provided upon request. The audio CD contains the prerecorded material for both movements as well as a realization of the complete piece.

Notes on *Saturn's Child*: Rhythmic notation (except in the dance section) is approximately spatial. Rhythmic freedom is desired. The alignment of toy piano to electronics is also somewhat flexible but should generally align as notated in the score. The recorded realization on the CD is meant only as a guide, not as a definitive interpretation.

Note on *Clockwise*: Kick it!!

Movements may be performed separately if desired.

Duration: ca. 10 min.

Please contact the composer regarding any performances:
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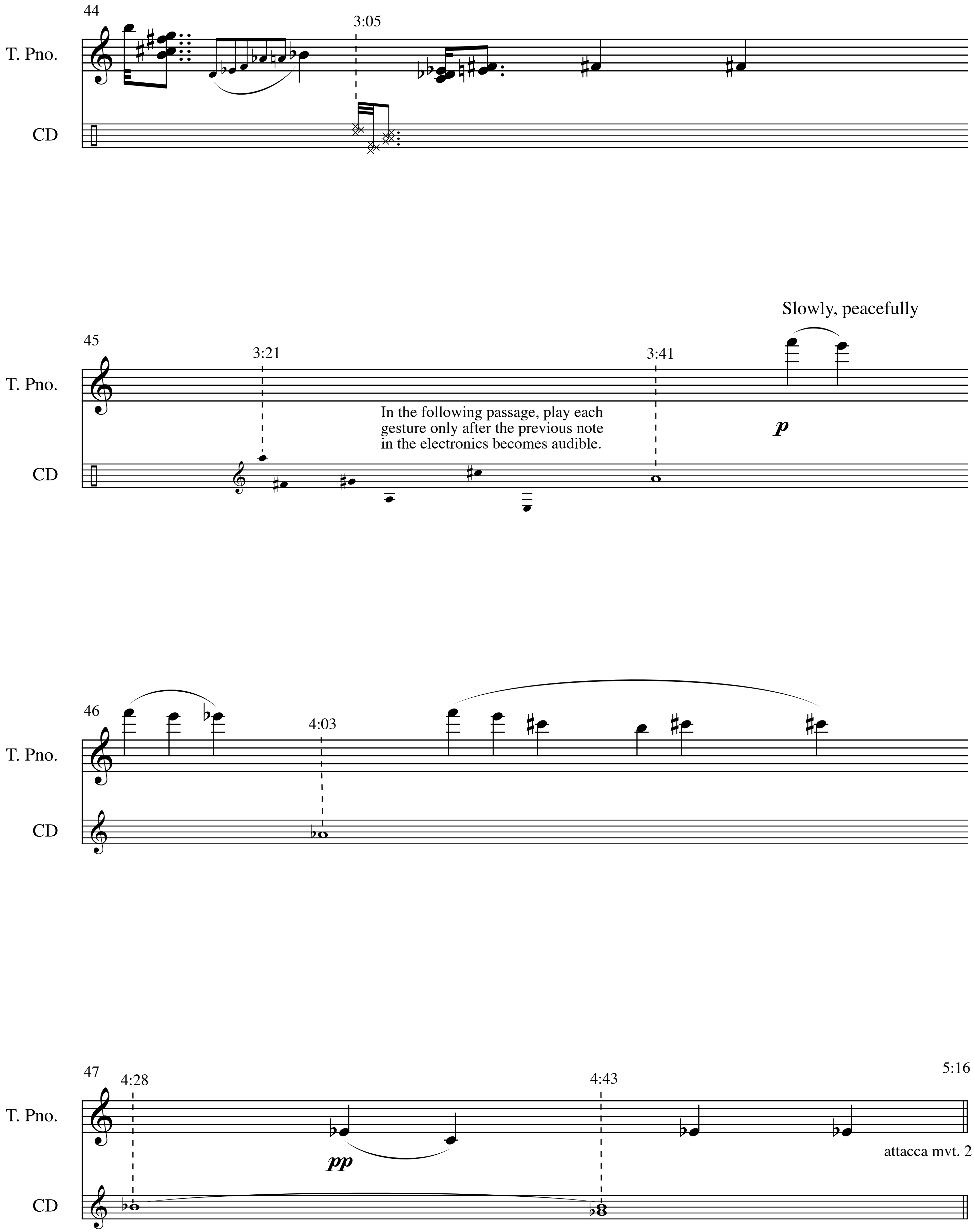
I. Saturn's Child

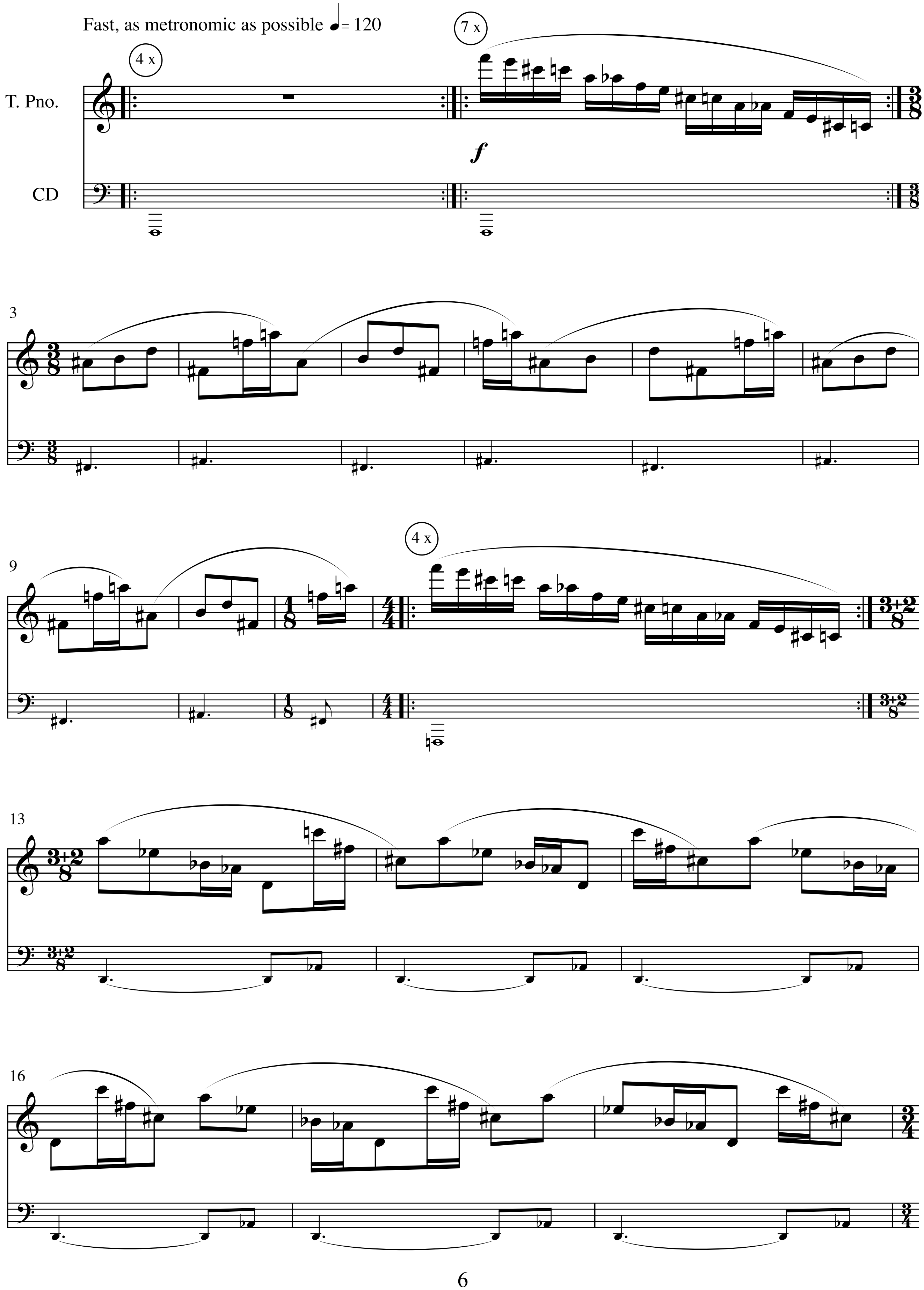

5
accidentals apply to repeated notes throughout the mvt.
T. Pno.
mf
6
1:49
1:53
2:06
f
p
p
microtonal chord
CD
7
Dance it! ♩= ca. 137.14
mf
junk metal dance
11
Faster ♩= 160
15


19
T. Pno.
CD
Faster! ♩= 192
23
T. Pno.
CD
add weirdvibe dance
27
T. Pno.
CD
Wild presto!! ♩= 240
31
T. Pno.
CD


35
T. Pno.
CD
39
T. Pno.
CD
2:50
42
T. Pno.
background sustain gets louder
CD
Violently!
2:56
3:00
43
T. Pno.
fff
* fully chromatic clusters
in the range indicated
CD


44
3:05
T. Pno.
CD
45
3:21
3:41
Slowly, peacefully
In the following passage, play each gesture only after the previous note in the electronics becomes audible.
p
46
4:03
47
4:28
4:43
5:16
pp
attacca mvt. 2

II. Clockwise




19
T. Pno.
CD
21
T. Pno.
CD
23
T. Pno.
CD
25
T. Pno.
CD
29
T. Pno.
CD

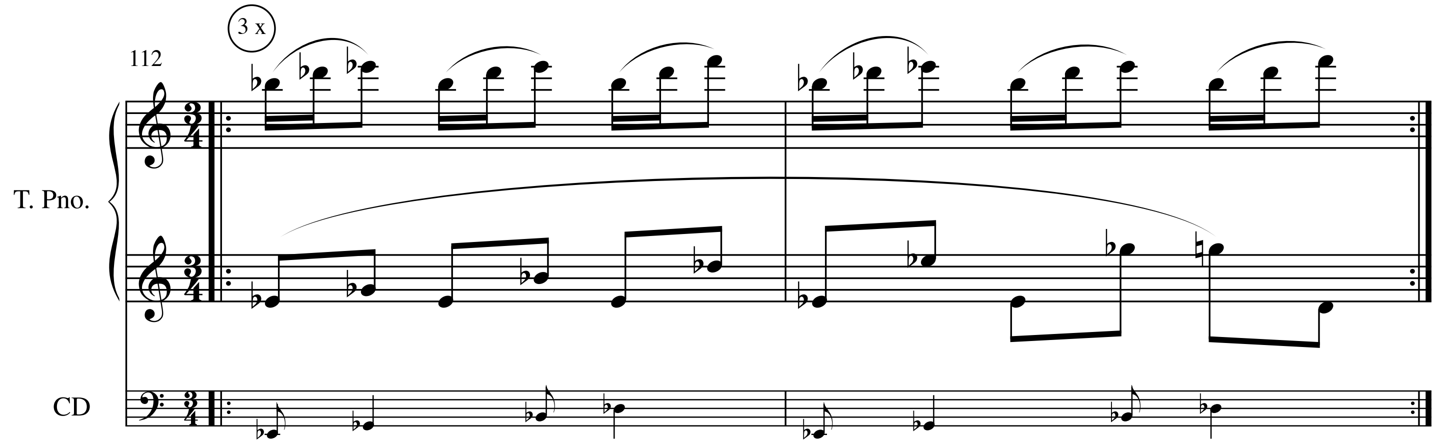

33
T. Pno.
CD
37
T. Pno.
CD
41
T. Pno.
CD
46
T. Pno.
CD
50
T. Pno.
CD


54
T. Pno.
CD
2 x
58
61
3 x
63
65

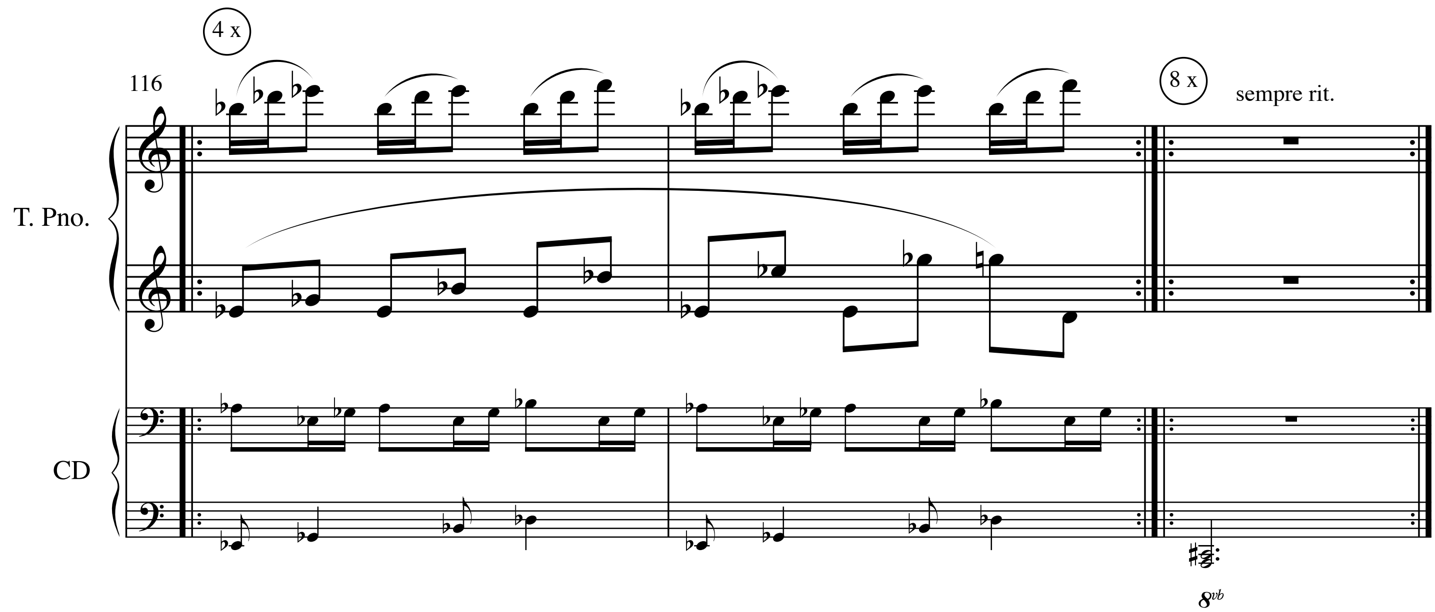

67
T. Pno.
CD
2 x
70
T. Pno.
spectral glissandi
CD
75
T. Pno.
CD
81
T. Pno.
CD


87
T. Pno.
CD
93
T. Pno.
CD
97
T. Pno.
CD
100
T. Pno.
CD


102
T. Pno.
CD
104
T. Pno.
CD
106
T. Pno.
CD
109
T. Pno.
CD


3 x
112
T. Pno.
CD


114
T. Pno.
CD


4 x
116
8 x
sempre rit.
T. Pno.
CD
8vb